Multi-Task Graph Adaptive Learning for Multivariate Electricity Price Short-Term Forecasting in Australia's National Electricity Market

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***Abstract*—Accurate electricity price short-term forecasting plays an essential role in the digitization of the electricity market. However, due to the expansion of renewable energy resources and the development of electricity demands, electricity prices are increasingly volatile and difficult to predict, posing a significant threat to the security of daily electricity market operations. The uncertainty of the supply-demand balance, the spatiotemporal correlation of the electricity market are two major obstacles to making the forecasting precisely. In this paper, a multi-task learning model (MGAAL) utilizes a graph attention mechanism and incorporates an auxiliary task focused on predicting abnormal price spikes, enhancing generalization and reducing overfitting risk. Specifically, MGAAL employs attention-based Graph Neural Networks to enhance price forecasting by capturing temporal and spatial power flow dynamics. In addition, MGAAL can also adaptively assign task weights based on homoscedasticity uncertainty and gradient normalization of the tasks. Finally, our experiments, conducted using data from Australia's National Electricity Market (NEM), demonstrate the effectiveness of MGAAL, surpassing current state-of-the-art methods.**

***Index Terms*—Anomaly event detection, electricity price forecasting, graph neural networks, multi-task learning, price spiking prediction.**

NOMENCLATURE

wd	The input sliding window with d width.
wh	The output forecasting window with h width.
$x_t^{i,RP}$	Electricity price of region i at time t
$x_t^{i,D}$	Electricity demand of region i at time t
$x_t^{i,G}$	Electricity generation of region i at time t
$x_t^{i,MW}$	Flow across of region i at time t
$x_t^{i,L}$	Region i price spikes label at time t
$x_t^{i,M}$	Region i 28-day average price at time t
$x_t^{i,B}$	Business features of region i at time t
$\hat{x}_t^{i,RP}$	Forecasting price of region i at time t
$\widetilde{x}_t^{i,RP}$	Reconstructed price of region i at time t
$\hat{x}_t^{i,L}$	Forecasting price label of region i at time t
$\widetilde{x}_t^{i,L}$	Reconstructed price label of region i at time t
$x_t^{i,f} = (x_t^{i,D}, x_t^{i,G}, x_t^{i,MW}, x_t^{i,M})^T$	External features of region i at time t
$X_t^f = (x_t^{1,f}, \ldots, x_t^{5,f})^T$	External features of all regions at time t
$X^f = \{X_1^f, \ldots, X_d^f\}$	External features of all regions in wd
$X_t^B = (x_t^{1,B}, \ldots, x_t^{5,B})$	Business features of all region at time t
$X^B = \{X_1^B, \ldots, X_d^B\}$	Business features of all regions in wd
$X_t^{RP} = (x_t^{1,RP}, \ldots, x_t^{5,RP})^T$	The real price of all regions at time t
$X_{in}^{i,RP} = (x_1^{i,RP}, \ldots, x_d^{i,RP})^T$	The real price of region i in wd
$\widetilde{X}_{in}^{i,RP} = (\widetilde{x}_1^{i,RP}, \ldots, \widetilde{x}_d^{i,RP})^T$	The reconstructed price of region i in wd
$X_{in}^{RP} = (X_{in}^{1,RP}, \ldots, X_{in}^{5,RP})^T$	The real price of all regions in wd
$X_{out}^{i,RP} = (x_1^{i,RP}, \ldots, x_h^{i,RP})^T$	The real price of region i in wh
$\hat{X}_{out}^{i,RP} = (\hat{x}_1^{i,RP}, \ldots, \hat{x}_h^{i,RP})^T$	The predicted price of region i in wh

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$X_t^L = (x_t^{1,L}, \ldots, x_t^{5,L})^T$ The event features of all regions at time t

$X_{in}^L = (X_1^L, \ldots, X_d^L)^T$ The label features of all regions in wd

$X_{in}^{i,L} = (x_1^{i,L}, \ldots, x_d^{i,L})^T$ The real event label of region i in wd

$\widetilde{X}_{in}^{i,L} = (\widetilde{x}_1^{i,L}, \ldots, \widetilde{x}_d^{i,L})^T$ The reconstructed event labels of region i in wd

$X_{out}^{i,L} = (x_1^{i,L}, \ldots, x_h^{i,L})^T$ The real event label of region i in wh

$\hat{X}_{out}^{i,L} = (\hat{x}_1^{i,L}, \ldots, \hat{x}_h^{i,L})^T$ The predicted event label of region i in wh

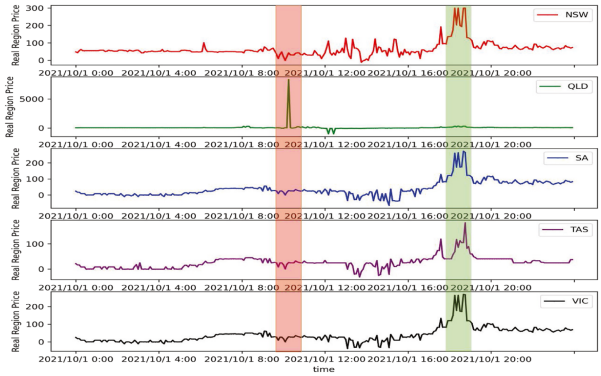


Fig. 1. Example of the electricity prices of NEM on 2021.10.01. Green indicates normal values and red indicates anomalies in QLD.

I. Introduction

ELECTRICITY is a unique commodity that can be purchased by retailers at unregulated spot prices and sold to consumers at heavily regulated prices [1]. As a result, fluctuations in spot electricity prices pose a significant risk to retailers' profits. Additionally, as prices become more volatile, it can compromise the stability of the grid and increase the risk of blackouts. In Australia, from October 2021 to January 2022, the spot prices of electricity experienced unusual volatility, with the average price nearly four times higher than the same period the previous year and reaching a high of $14,000/MWh at its peak [2]. These price spikes caused economic losses for market participants and disrupted the development of the electricity market. Therefore, it is crucial to precisely short-term forecast electricity prices for effective risk management in electricity markets, as this not only has the potential to generate economic profits but also improves the power system stability.

Electricity price short-term forecasting has long been at the center of intensive studies. Traditional approaches, for example, ARIMA-based models and ensemble learning models have been utilized in electricity price forecasting [3], [4], [5] in electricity market. Recently, deep learning methods have gained attention, paper provided a noising autoencoder based on DNN to forecast the price [6]. Some researches are based on LSTM models for electricity price forecasting [7], [8], [9], [10], [11]. Electricity prices can be influenced by various factors such as local demand and supply imbalance, as well as the influence of neighboring areas [12]. Papers [13], [14] consider features from connected markets to improve predictive accuracy. Additionally, some papers examine price spikes as anomalies [15], [16]. Paper [15] develops a multi-feature based approach incorporating variable thresholds for detecting price spikes. However, the task of short-term electricity price forecasting still poses several challenges.

One challenge is the uncertainty of rare price spike events [17] where the sudden rise in electricity prices can be caused by unexpected demands, supply shortages, and transmission failures. Given that uncertainty in electricity prices can significantly impact price forecasting and create market uncertainty [18]. Accurately predicting these spikes is critical for effective risk management in the retail energy sector [1]. While previous research has attempted to improve the accuracy of electricity price forecasting [19], but neglected to pay attention to unusual electricity price spike events. Paper [20] proposes a two-step day-ahead electricity price forecasting algorithm by modeling price spike events. Also, the paper [21] proposes a model that includes an outlier filter to improve the accuracy of the price prediction values. However, features of electricity prices captured by a single-task learning model are often insufficient to comprehensively and accurately reflect its dynamic change [22]. In [23], a combined prediction model based on multi-task learning and least square support machine can accurately predict the power, heat, cooling, and gas loads of integrated energy systems. Therefore, it is natural to think of predicting electricity price spike events as an auxiliary task for improving the performance of price forecasting.

The second challenge is that it is difficult to accurately forecast electricity prices due to the dynamic nature of integrating renewable energy resources and the electricity demands in terms of both temporal and spatial dimensions [24]. These dynamics can have a significant impact on electricity prices in different locations. Historical observations from the same location can also have different impacts on future price trends at different time slots [25]. This is illustrated in Fig. 1, which shows a boost in price in all regions during the green segment, which is normal, but an inconsistent pattern in QLD during the red segment, which is abnormal. It is therefore important to consider the correlations between different time series to accurately forecast multivariate electricity prices. The existing research has attempted to incorporate the entire wholesale market in their electricity price forecasting [26]. However, these approaches do not explicitly capture correlations between different markets [27]. Graphs are a special form of data that describes the relationships between different entities. By using GNNs [28], [29], [30], it is possible to analyze the relationships between different time series and make more accurate predictions about future electricity prices. Several studies have employed Graph Neural Networks (GNNs) for electricity forecasting. For instance, paper [31] proposes a Chebyshev graph neural network to predict the optimal marginal market price in a real-time electricity market environment. The paper [32] utilizes graph neural networks to predict prices and energy flows in the context of spatiotemporal relationships in energy markets. Inspired by these studies, we aim to develop a novel method leveraging GNN to effectively capture the

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spatiotemporal characteristics inherent in electricity prices within electricity markets.

To address these challenging issues in electricity price forecasting, our research mainly focuses on the short-term forecasting problem in Australia's NEM which is formed by the regional markets of New South Wales (NSW), Queensland (QLD), Victoria (VIC), South Australia (SA), and Tasmania (TAS). In addition, we will conduct multi-step ahead forecasting, which is essential for power market trading and provides more informative output for decision-makers to optimize their actions in case of rare events. The major contributions are highlighted as follows:

1) We propose a novel Graph Neural Networks (GNN)-based framework to forecast electricity prices. Neighboring electricity markets are modeled as graph nodes, and their time-series spatial correlations are captured using GNNs, while dependencies in the time dimension are learned using an LSTM-based encoder-decoder model.
2) MGAAL is modeled as a multi-task optimization method. By adding auxiliary tasks, optimization objectives (event prediction loss), and simultaneously considering electricity price forecasting and anomaly price spiking prediction tasks, we can increase the amount of data to train. Therefore, the model generalization capability is boosted by learning temporal features from different perspectives.
3) When dealing with price forecasting loss and price spiking anomaly prediction loss, the loss function of the multi-task is combined using homoscedasticity uncertainty as well as gradient normalization to learn multiple objectives simultaneously, achieving adaptive assignment of task weights according to the speed of gradient descent.
4) Due to the complicated nature of MGAAL, it may struggle to capture long-range dependencies in underlying graphs which could lead to difficulty in fine-tuning the hyperparameters. We prove the convergence of this tensor spatiotemporal graph neural network, and this convergence proof can help to tune the hyperparameters which can be beneficial for improving the performance of the proposed approach.

The rest of the paper is organized as follows: Dataset description and problem formalization are introduced in Section II. The overall framework of the proposed MGAAL is described in detail in Section III, and the convergence of this tensor spatiotemporal graph neural network and optimization in adaptive weighting are introduced in detail. Comparative experiments and analysis are conducted on the Australian dataset in Section IV. Finally, conclusions for this article are provided.

II. Preliminaries

In this section, we first introduce the collected dataset used in this paper and the problem will be restated.

A. Dataset and Notation Description

The dataset was collected from October, 1, 2021 to January, 31, 2022 according to the NEM web site [2], which consists of 5 states data. For each area $i \in \{1, 2, 3, 4, 5\}$ (1:*NSW*, 2:*QLD*, 3:*SA*, 4:*TAS*, 5:*VIC*), not only alongside electricity price $x_t^{i,RP}$ at time t, this dataset includes external features such as demand $x_t^{i,D}$, generation $x_t^{i,G}$, power flow $x_t^{i,MW}$, Region i 28-day average price $x_t^{i,M}$ at time t, and time feature. For convenience, all subscripts t in this paper denote values located at moment t and $x_t^{i,f} = \{x_t^{i,D}, x_t^{i,G}, x_t^{i,MW}, x_t^{i,M}\}$. The temporal resolution of the dataset is 5 min, the dataset comprises a total of 35,424 data points. To evaluate anomaly events, labels of region i at time t ($x_t^{i,L}$) have been assigned in the following manner.

$$\begin{cases} x_t^{i,L} = 2, x_t^{i,RP} > 2 * x_t^{i,M}, \\ x_t^{i,L} = 0, x_t^{i,RP} < 0.5 * x_t^{i,M}, \\ x_t^{i,L} = 1, \text{others}, \end{cases} \tag{1}$$

where $x_t^{i,L} = 2$ stands for high level anomaly price spikes and $x_t^{i,L} = 0$ stands for low level anomaly price. Spike occurrences in electricity prices for various states are as follows: NSW(6.21%), QLD(12.66%), SA(29.69%), TAS(20.29%), VIC(26.17%). Notably, the thresholds are derived from a combination of labeling methods in anomaly detection and examination of the dataset, which is a common practice in the field of labeling abnormal electricity prices [15]. Moreover, the threshold is adaptive to different markets at different times.

B. Feature Engineering

Before feeding the data into the model, feature engineering is conducted on the time series to extract the hidden latent variables. We generate the business feature $x_t^{i,B}$ for a given time that particular hour and weekend which it falls into. Weekend electricity prices are typically lower compared to weekdays, while nighttime prices tend to be higher than daytime prices. As a result, we assign a 'business hour' value of '1' for hours falling within the business hours range (17:00–23:00), and '0' for all other hours. Likewise, the 'business week' value is '1' for weekend days (Saturday and Sunday) and '0' for all other weekdays, reflecting the distinctive pricing patterns.

C. Problem Statement

This research aims to predict not only electricity prices but also different categories of anomalies in each state, to manage the electricity market in a more timely and efficient manner. We transformed the price spiking prediction task into a supervised anomaly detection problem by labeling the dataset. A multi-task learning model that simultaneously predicts electricity prices and price spike events in order to improve the accuracy of the main task regarding price forecasting.

With respect to the forecasting model, our target is to outperform h-steps by using input sliding windows d for both electricity prices and abnormal price spikes. The target can be present as follows:

$$f : \{X_{in}^{RP}, X^f, X_{in}^L, X^B\} \rightarrow \{\hat{X}^{RP}, \hat{X}^L\} \tag{2}$$

where X_{in}^{RP}, X^f, X_{in}^L and X^B denote the real price, external features, anomaly labels and business features of all regions in wd. While characters with sharp corner ($\hat{A}$) indicate predicted values of A in h ahead steps.

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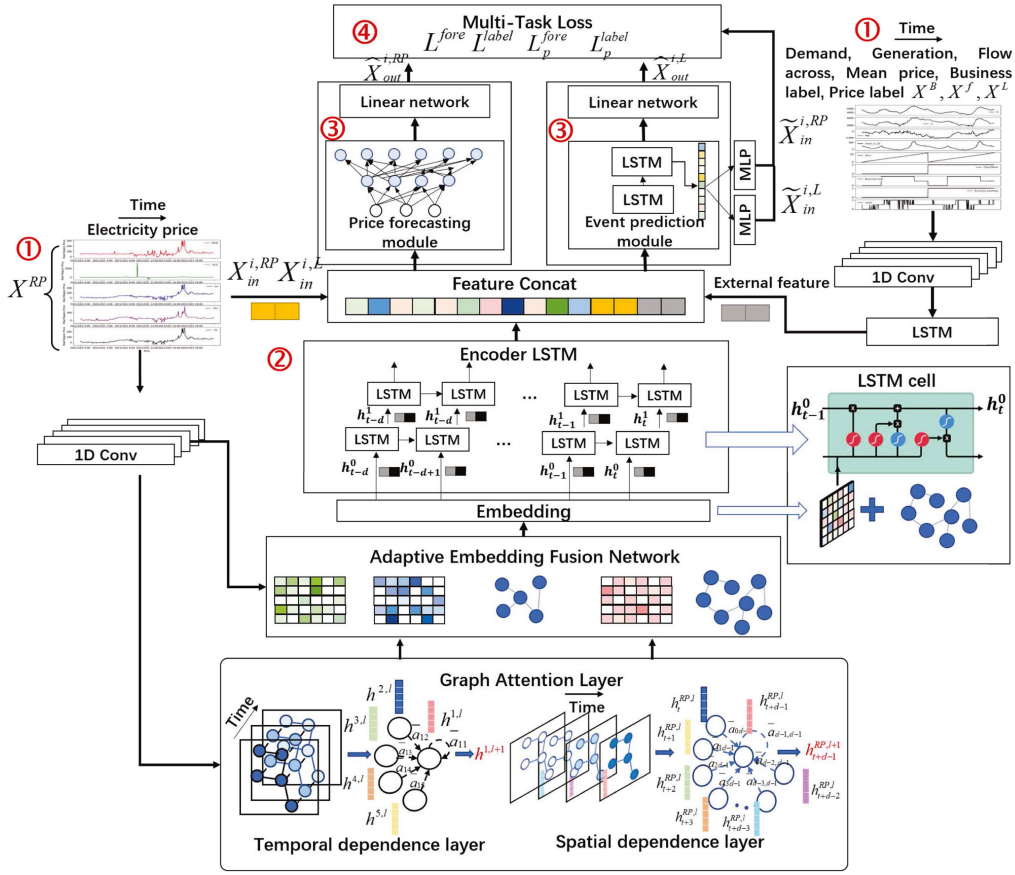


Fig. 2. Architecture of MGAAL. (1). Input: This framework has two input sources. (2). Feature extractor: The features from 1D Conv and Graph Attention Module modules will be adaptive fused through an embedding fusion network and then input to the LSTM encoder to capture long-range dependencies. Then the main and external features will be concatenated with a residual structure, and input to the forecasting modules. (3). Forecasting module: These modules serve two purposes: predicting prices and anomaly price spikes and reconstructing past prices and spikes. (4). Optimization module: The forecasting losses and reconstructed losses will be combined to optimize the network.

III. Methodology

This section will describe the framework of the MGAAL model in detail, including the graph attention layer, the multi-task learning loss optimization, and the forecasting modules for the two tasks.

A. Architecture of MGAAL

The main idea of the model is to combine the main and external features to predict electricity prices. We consider a hybrid structure to combine main features coding and external features coding. This hybrid structure consists of a graph neural network layer, which learns and models the relationship between the main features (X_{in}^{RP}) in the graph, and a recurrent layer, which learns and models the sequential relationship between the external features in the time series (X^B, X^f, X_{in}^L). The electricity prices, as the main features that need to be predicted, are affected by spatiotemporal factors, so the past time series is input into the GNN layer. The external features are used as auxiliary information for the electricity prices, without considering the influence of other markets, and are processed through the recurrent layer. The resulting embeddings from both feature extractors are then combined and input into the forecasting module. The overall framework of the MGAAL model is shown in Fig. 2.

1) Input: There are two input sources in the framework, electricity prices X_{in}^{RP}, and external features, X^B, X^f, X_{in}^L.

2) Feature Extractor: The MGAAL model first uses 1D Conv layer to extract high-level features from the input electricity prices time-series data. These features then undergo processing in a graph neural network (GNN) layer, enabling the model to capture relationships across different time series in both spatial and temporal dimensions. The features obtained from these three modules are intelligently combined through a linear network. Subsequently, an encoder, comprising LSTM layers, is employed to capture long-term dependencies within the data in the graph. While external features serve as auxiliary inputs and follow a separate processing path. They are input to 1D Conv layer and a recurrent layer utilizing LSTM to capture sequential relationships among these external features within the time series data. Importantly, these external features are not integrated into the GNN layer, instead, they are used to provide supplementary context for this specific forecasting task. To prevent degradation, we consider the residual structure, specifically, when concatenating the main features and external features, we concatenate both $X_{in}^{i,RP} = (x_1^{i,RP}, \ldots, x_d^{i,RP})^T$

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and $X_{in}^{i,L} = (x_1^{i,L}, \ldots, x_d^{i,L})^T$. The high-dimensional concatenating features will be input to the downstream tasks.

3) Forecasting Module: There are 2 forecasting modules as the downstream tasks. One is a prediction decoder composed of FNN for electricity price forecasting and the other is an anomaly price spiking prediction decoder composed of LSTM.

4) Optimization Module: The power of the price forecasting module and the event prediction module is exploited by optimizing a joint objective function. In this case, the loss consists of five components: prediction loss, event prediction loss, intermediate supervised loss, encoder-decoder past value loss, and encoder-decoder past label loss.

The following of this section is organized as follows. The details of embedding obtained, graph attention layers will be presented in Section III-B. The forecasting module and optimization module are described in Sections III-C and III-D, respectively.

B. Graph Attention Layer

The graph attention layer is based on the message-passing mechanism of GNN for node embedding updates. The attention mechanism is utilized when node features are aggregated from neighboring nodes, thus different weights are considered for learning the relationship between nodes. There are two different graph attention layers in MGAAL model. 1). Spatial dependence layer, 2).Temporal dependence layer.

The spatial dependence layer applies to extract the relationship between different markets, which can be seen at the bottom of Fig. 2. With d width sliding window, the electricity prices time sequence of different markets will be modeled as nodes in the graph G. And $h^{i,l+1} \in R^{f^{l+1}}$ represents the output of $l+1$th spatial dependence layer of node i, where f^{l+1} represents the dimensions of the extracted features. The weight of the GAT layer is computed by a self-attention mechanism based on node features. Where the attention $\overline{a}_{ij}$ between node i and node j can be calculated as follows [33]:

$$\overline{a}_{ij} = \frac{exp(a_{ij})}{\sum_{k=1}^{N-1}(a_{ik})}, a_{ij} = LeakyReLU(W_a^T \cdot (h^{i,l} \| h^{j,l})), \tag{3}$$

where $W_a^T \in R^{2\,d}$ denotes the learnable weight, which can convert the input features to high-dimensional features (incremental dimension) to obtain sufficient expressiveness. After obtaining the normalized attention coefficients, the output of lth spatial dependence layer of node i can be updated as follows:

$$h^{i,l+1} = \sigma\left\{\sum_{j \in N(i)} a_i^l W^l h^{j,l}\right\}, \tag{4}$$

where $N(i)$ denotes the other node of this graph, W^l is learnable weight matrix of lth layer and $h^{i,0} = x_t^{i,RP}$. Therefore, the embedding of nodes and spatial dependence can be learned.

Similarly, at the temporal dependence layer, unlike the spatial dependency layer, each timestamp of all nodes in d width sliding window will be contacted together and treated as a graph node, which also can be seen at the bottom of Fig. 2. There are d nodes in the graph with the initial embedding $X_t^{RP} = (x_t^{1,RP}, \ldots, x_t^{5,RP})^T$. The attention between nodes can be calculated like above, so there is no further discussion. Finally, the embedding of spatial dependence nodes and temporal dependence nodes can be obtained and adaptively fused with the features extracted in 1D Conv as the spatial-temporal embedding and the next module input.

C. Tensor-Based Spatial-Temporal Graph Ordinary Differential Equation(ODE)

In this subsection, we will prove some properties of the model and the convergence of the model when the conditions are satisfied. In this paper, in order not to lose generality, the variable is in Spatial-Temporal tensor form: $X_{in}^{RP} = (X_{in}^{1,RP}, \ldots, X_{in}^{k,RP})^T \in R^{k \times d}$.$k$ denotes the number of electricity markets. Tensor has some multiplication methods, and the following multiplication method will be used in this section. Assume $\mathcal{T} \in R^{n_1,n_2,n_2}$, $\mathcal{M} \in R^{n_1,n_2,n_3}$, then $\mathcal{T} \otimes \mathcal{M} \in R^{n_1,n_2,n_3}$, where $\otimes$ denotes tensor multiplication with broadcast mechanism.

For hidden variables, $h_c^{i,l+1} = \sigma(\sum W_c^l * h^{j,l})$ denotes the output of 1D Conv layer, $h_s^{i,l+1} = \sigma(\sum_{j \in N(i)} a_{ij,s}^l W_s^l h^{j,l})$ denotes the output of spatial dependence layer and $h_t^{i,l+1} = \sigma(\sum_{j \in N(i)} a_{ij,t}^l W_t^l h^{j,l})$ denotes the output of temporal dependence layer.

$$H_s^l = \left\{\underbrace{h_s^{1,l}, \ldots, h_s^{k,l}}_{\text{batch 1}}, \ldots, \underbrace{h_s^{1,l}, \ldots, h_s^{k,l}}_{\text{batch b}}\right\} \in R^{b \times k \times f^l}, \tag{5}$$

$$H_t^l = \left\{\underbrace{h_t^{1,l}, \ldots, h_t^{k,l}}_{\text{batch 1}}, \ldots, \underbrace{h_t^{1,l}, \ldots, h_t^{k,l}}_{\text{batch b}}\right\} \in R^{b \times f^l \times k}, \tag{6}$$

$$H_c^l = \left\{\underbrace{h_c^{1,l}, \ldots, h_c^{k,l}}_{\text{batch 1}}, \ldots, \underbrace{h_c^{1,l}, \ldots, h_c^{k,l}}_{\text{batch b}}\right\} \in R^{b \times k \times f^l}, \tag{7}$$

where b represents the batch number for training, $a_{ij,s}^l$, $a_{ij,t}^l$ denote the spatial attention score and temporal attention score between nodes i and j, respectively.

The spatial attention matrix of the spatial dependence layer can be denoted as $A_s^l \in R^{b \times k \times k}$, while the temporal attention matrix of the temporal dependence layer can be denoted as $A_t^l \in R^{b \times f^{l+1} \times f^l}$. In addition, the learning matrix of spatial dependence layer $W_s^l \in R^{f^l \times f^{l+1}}$, the learning matrix of temporal dependence layer $W_t^l \in R^{k \times k}$. $W_c^l \in R^{k \times k_c}$ denotes 1D convolution kernel, k_c denotes the kernel size, $*$ represents convolution operation. In this model, we define $f^{l+1} = f^l = f^0 = d$. Then,

$$H_c^{l+1} = \sigma(W_c^l * H_c^l) \in R^{b \times d \times k}, \tag{8}$$

$$H_s^{l+1} = \sigma(A_s^l \otimes H_s^l \otimes W_s^l) \in R^{b \times k \times d}, \tag{9}$$

$$H_t^{l+1} = \sigma(A_t^l \otimes H_t^l \otimes W_t^l) \in R^{b \times d \times k}. \tag{10}$$

The final hidden output will be adaptively fused as

$$H^{l+1} = \alpha^l H_c^{l+1} W_\alpha^l + \beta^l H_{s,(2,3)}^{l+1} W_\beta^l + \gamma^l H_t^{l+1} W_\gamma^l, \tag{11}$$

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where (2,3) denotes exchange the order of the 2nd, 3 rd dimensions of the tensor. The 3 features are first extracted by 3 linear neural networks($W_\alpha^l, W_\beta^l, W_\gamma^l \in R^{k\times d}$), and then fused by α^l, β^l, γ^l adaptively. α^l, β^l, γ^l are the weights of the three features, and $\alpha^l + \beta^l + \gamma^l = 1$. These three parameters can be learned adaptively by the network and defined: $\alpha^l = \frac{e^{\alpha^l}}{e^{\alpha^l}+e^{\beta^l}+e^{\gamma^l}}$, the other two parameters are defined in the same way. The discrete formulation is extended to a continuous expression. Intuitively, l is replaced with a continuous variable t. The model is deformed to get

$$H^{l+1} - H^l = -H^t + \alpha^l H_c^{l+1} W_\alpha^l + \beta^l H_{s,(2,3)}^{l+1} W_\beta^l + \gamma^l H_t^{l+1} W_\gamma^l.$$

For convenience, $F(t) = \alpha(t)\sigma(W_c(t) * H_c(t))W_\alpha(t) + \beta(t)\sigma(A_s(t) \otimes H_s(t) \otimes W_s(t))_{(2,3)}W_\beta(t) + \gamma(t)\sigma(A_t(t) \otimes H_t(t) \otimes W_t(t))W_\gamma(t)$. Thus, the continuous MGAAL model is given by

$$\dot{H}(t) = -H(t) + F(t), \tag{12}$$

which is governed by the differential equation.

Lemma 1: The analytic solution of (12) is given by

$$H(t) = e^{-t}e^{t_0}H(t_0) + e^{-t}\int_{t_0}^{t} e^x F(x)dx \tag{13}$$

Proof: By mathematical concepts of the differential equation, it yields

$$\dot{H}(t) + H(t) = F(t), \tag{14}$$

$$\frac{d\left(e^t \dot{H}(t)\right)}{dt} = e^t F(t). \tag{15}$$

$$H(t) = e^{-t}e^{t_0}H(t_0) + e^{-t}\int_{t_0}^{t} e^x F(x)dx \tag{16}$$

□

Lemma 2: If elements in $H(t_0) \in [-1, 1]$, then elements in $H(t) \in [-\alpha W_\alpha - \beta W_\beta - \gamma W_\gamma, \alpha W_\alpha + \beta W_\beta + \gamma W_\gamma]$, where $t \geq t_0 \geq 0$.

Proof: According to Lemma 1

$$H(t) = e^{-t}e^{t_0}H(t_0) + (1 - e^{-t}e^{t_0})\int_{t_0}^{t} \frac{e^x}{e^t - e^{t_0}} F(x)dx. \tag{17}$$

Because $\frac{e^x}{e^t - e^{t_0}} > 0$ and $\sigma(\cdot) = \tanh(\cdot)$, $\sigma(W_c * H_c(t)) \in [-1, 1], \sigma(A_s \otimes H_s(t) \otimes W_s)_{(2,3)}) \in [-1, 1], \sigma(A_t \otimes H_t(t) \otimes W_t) \in [-1, 1]$, for $t_0 \leq x \leq t$, $\int_{t_0}^{t} \frac{e^x}{e^t - e^{t_0} dx} = 1$. Note that the $[-1, 1]$ is a convex set.

$$(1 - e^{-t}e^{t_0})\int_{t_0}^{t} \frac{e^x}{e^t - e^{t_0}} F(x)dx \in [-\alpha W_\alpha - \beta W_\beta - \gamma W_\gamma, \alpha W_\alpha + \beta W_\beta + \gamma W_\gamma] \tag{18}$$

According to the definition of convex set, the elements in $H(t) \in [-\alpha W_\alpha - \beta W_\beta - \gamma W_\gamma, \alpha W_\alpha + \beta W_\beta + \gamma W_\gamma]$, where $t \geq t_0 \geq 0$. □

Theorem 1: The continuous update *H(t)* (13) is convergent if the $F(t)$ satisfies $\|F(t) - H^*\| \leq \|H(t) - H^*\|$

Proof: In the iterative process, $\alpha(t)$, $\beta(t)$, $\gamma(t)$, $W_\alpha(t)$, $W_\beta(t)$, $W_\gamma(t)$, $W_s(t)$, $W_t(t)$, $W_c(t)$, $A_s(t)$, $A_t(t)$ can be regarded as constants, so $H(t)$ satisfies the Lipschitz condition.

In order to prove the convergence of the update method $H(t)$, let the Lyapunov function defined by $V(t) = \frac{1}{2}\|H(t) - H^*\|^2$, where H^* is the equilibrium point of $H(t)$, H^* represents the ideal solution of the steady state of the model. Then,

$$\begin{aligned}\dot{V}(t) &= (H(t) - H^*)^T \dot{H}(t) \\ &= (H(t) - H^*)^T (H^* - H(t) + F(t) - H^*) \\ &= -\|H^* - H(t)\|^2 + (H(t) - H^*)^T (F(t) - H^*).\end{aligned} \tag{19}$$

According to the Lipschitz condition of $\sigma(\cdot)$, Lemma 1 and 2, one has $H^* = F^*(t)$ if $\|F(t) - H^*\| \leq \|H(t) - H^*\|$ is satisfied. Thus, $\dot{V}(t) \leq 0$. The proof is completed. □

D. Price Forecasting Module

For multi-task learning, there are two forecasting modules: 1) The price forecasting module. 2) The price spiking forecasting module. Both modules are designed using an encoder-decoder framework. The encoder-decoder framework is not only used to predict prices $\hat{x}_t^{i,RP}$ and labels $\hat{x}_t^{i,L}$ but also to reconstruct past prices $\widetilde{x}_t^{i,RP}$ and labels $\widetilde{x}_t^{i,L}$. In addition, the proposed MGAAL model uses a residual structure to improve the prediction of electricity prices and anomaly spike events by considering concatenation with $X_{in}^{i,RP}$ and $X_{in}^{i,L}$, which can be seen in Fig. 2.

The price forecasting module used multi-output forecasting. Namely, by the hidden layer output, the fully-connected network is used to forecast h steps ahead prices and reconstruct d steps past electricity prices. The price forecast loss function and past price reconstruct loss function of node i can be formulated as:

$$L_t^{i,fore} = \|\hat{x}_t^{i,RP} - x_t^{i,RP}\|^2, L_{p,t}^{i,fore} = \|\widetilde{x}_t^{i,RP} - x_t^{i,RP}\|^2. \tag{20}$$

For the price spiking prediction module, the anomaly event forecast loss function and anomaly event reconstruct loss function of node i can be calculated using the cross-entropy loss function:

$$L_t^{i,label} = \sum_{t=0}^{h-1}\left(-x_t^{i,L}\log\left(Softmax\left(\hat{x}_t^{i,L}\right)\right)\right), \tag{21}$$

$$L_{p,t}^{i,label} = \sum_{t=0}^{d-1}\left(-x_t^{i,L}\log\left(Softmax\left(\widetilde{x}_t^{i,L}\right)\right)\right). \tag{22}$$

Remark 1: In contrast to prior studies using encoder-decoder methods for anomaly detection, this research grapples with non-normally distributed training data. Consequently, the conventional approach of modeling data via normal distributions and detecting anomalies through high reconstruction errors isn't feasible. This study's aim is to predict both regular and abnormal electricity prices and events while ensuring robust model generalization. To achieve this, the proposed MGAAL model doesn't just predict electricity prices and labels but also reconstructs

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abnormal electricity prices and labels. This approach allows the model to learn the relationship between normal and abnormal electricity prices and labels.

E. Optimization of Multitask Learning Loss

According to the above, there exist four task losses for each state. $L^{i,fore}$, $L^{i,label}$, $L_{p,t}^{i,fore}$ and $L_{p,t}^{i,label}$. Moreover, in this case, the total loss additionally includes the loss of intermediate supervisory loss L^{mean_fore} for the regularization term. To obtain the intermediate supervisory loss, we first compare the price forecasting output with the average value of that moment over the past 28 days to obtain the corresponding label $\overline{x}_t^{i,L}$ through process 1. Then, we calculate the cross-entropy loss by comparing this forecast value with the target label.

$$L_t^{mean_fore} = \sum_{t=0}^{h-1} \left(-x_t^{i,L} \log \left(Softmax \left(\overline{x}_t^{i,L} \right) \right) \right). \quad (23)$$

Therefore, the total loss:

$$L_t^i = w_t^{label} L_t^{i,fore} + w_t^{label} L_t^{i,label} + w_{p,t}^{i,label} L_{p,t}^{i,fore} + w_{p,t}^{i,label} L_{p,t}^{i,label} + L_t^{mean_fore}. \quad (24)$$

Tuning these weights by hand is a difficult and expensive process, making multi-task learning prohibitive in practice. Given the above, we aim to learn "w" with the following goals: 1). Adaptive weights, vary continuously with training step t and are learnable parameters. 2). Balancing the learning rates of different tasks. That is because, when one task learns too fast, it is possible that the other task learning will have less impact on the total loss.

Therefore, we apply the adaptive weight learning method by paper [34]. This paper proposes a principled way of deriving a multi-task loss function based on maximizing the Gaussian likelihood with homoscedastic uncertainty. Suppose $f^w(x)$ stands for the output of the forecast module with weights $\mathbf{W}$ on input x. For the price forecasting task, the uncertainty can be modeled as $p(\hat{y}_1|f^{\mathbf{W}}(x)) = N(f^{\mathbf{W}}(x), \sigma_1^2))$. While price spiking event prediction task, the uncertainty can be modeled as $p(\hat{y}_2|f^{\mathbf{W}}(x)) = Softmax(f^{\mathbf{W}}(x), \sigma_2^2)$. Similarly, the uncertainty of reconstructed price can be modeled as $p(\widetilde{y}_3|f^{\mathbf{W}}(x)) = N(f^{\mathbf{W}}(x), \sigma_3^2))$, and the uncertainty of reconstructed label can be modeled as $p(\widetilde{y}_4|f^{\mathbf{W}}(x)) = Softmax(f^{\mathbf{W}}(x), \sigma_4^2)$. σ_1, σ_2, σ_3, σ_4 are the learnable parameter, which this paper considers as the uncertainty corresponding to the task modeling. Given the above, we can model the total loss of region i as follows:

$$\begin{aligned}
&-\log p\left(X_{out}^{i,RP}, X_{out}^{i,L} = c_1 | f^w(x), X_{in}^{i,RP}, X_{in}^{i,L} = c_2 | f^w(x)\right) \\
&= -\log N\left(X_{out}^{i,RP}; f^{\mathbf{W}}(x), \sigma_1^2\right) \\
&\quad \cdot Softmax\left(X_{out}^{i,L} = c_1; f^{\mathbf{W}}(x), \sigma_2^2\right) \\
&\quad \cdot \log N\left(X_{in}^{i,RP}; f^{\mathbf{W}}(x), \sigma_1^2\right) \\
&\quad \cdot Softmax\left(X_{in}^{i,L} = c_2; f^{\mathbf{W}}(x), \sigma_2^2\right) \\
&= \frac{1}{2\sigma_1^2} \|X_{out}^{i,RP} - f^{\mathbf{W}}(x)\|^2 + \log \sigma_1 - \log p(y_2 = c_1 | f^{\mathbf{W}}(x), \sigma_2) \\
&+ \frac{1}{2\sigma_3^2} \|X_{in}^{i,RP} - f^{\mathbf{W}}(x)\|^2 + \log \sigma_3 - \log p(y_2 = c_2 | f^{\mathbf{W}}(x), \sigma_4) \\
&\approx \frac{1}{2\sigma_1^2} L_1(\mathbf{W}, t) + \frac{1}{\sigma_2^2} L_2(\mathbf{W}, t) + \log \sigma_1 + \log \sigma_2 \\
&+ \frac{1}{2\sigma_3^2} L_3(\mathbf{W}, t) + \frac{1}{\sigma_2^2} L_4(\mathbf{W}, t) + \log \sigma_3 + \log \sigma_4, \quad (25)
\end{aligned}$$

where $X_{in}^{i,RP}$ and $X_{in}^{i,L}$ denote the real price and real label of region i in sliding window d, $X_{out}^{i,RP}$ and $X_{out}^{i,L}$ denote the real price and real label of region i in sliding window h. $L_1(\mathbf{W}, t) = \|X_{out}^{i,RP} - f^{\mathbf{W}}(x)\|^2$, $L_3(\mathbf{W}, t) = \|X_{in}^{i,RP} - f^{\mathbf{W}}(x)\|^2$ and $L_2(\mathbf{W}, t) = -\log Softmax(X_{out}^{i,L}, f^{\mathbf{W}}(x))$ for the loss of $X_{out}^{i,L}$, $L_4(\mathbf{W}, t) = -\log Softmax(X_{in}^{i,L}, f^{\mathbf{W}}(x))$ for the loss of $X_{in}^{i,L}$. And optimise with respect to $\mathbf{W}$ as well as σ_1, σ_2, σ_3, σ_4. The total loss penalizes the task with a large loss and small σ, because for this task, the term will be large and the optimizer will optimize it to be small. To avoid the model to "taking a big step" in the wrong direction, w is updated with a smaller gradient. On the contrary, for the task with a small loss, its uncertainty is also low, and to update w with a larger gradient.

The above method solves the problem of adaptive assignment of task weights well, but it does not consider normalizing the losses of different tasks. Therefore, in this paper, the following normalization is performed.

$$\overline{L}_t^{i,fore} = L_1(\mathbf{W}, t)/L_1(\mathbf{W}, 0), \overline{L}_t^{i,label} = L_2(\mathbf{W}, t)/L_2(\mathbf{W}, 0).$$

$$\overline{L}_{p,t}^{i,fore} = L_3(\mathbf{W}, t)/L_3(\mathbf{W}, 0), \overline{L}_{p,t}^{i,label} = L_4(\mathbf{W}, t)/L_4(\mathbf{W}, 0).$$

$$\overline{L}^t = \frac{1}{2\sigma_1^2} \overline{L}_t^{i,fore} + \frac{1}{\sigma_2^2} \overline{L}_t^{i,label} + \frac{1}{2\sigma_1^2} \overline{L}_{p,t}^{i,fore} + \frac{1}{\sigma_2^2} \overline{L}_{p,t}^{i,label} + \log \sigma_1 \sigma_2 \sigma_3 \sigma_4, \quad (26)$$

where σ_i is the learning variable, $L_i(\mathbf{W}, 0)$ denotes the initial loss of $L_i(\mathbf{W})$.

IV. Experiment

In this section, we will proceed with several experiments for MGAAL on Australia datasets. First, we compare MGAAL model with some baseline models, moreover, an ablation study, external feature analysis, robustness experiment with different time periods and some visualization works will be done.

In the MGAAL model, the size of the sliding window is set as 64, the kernel size of 1D Conv is set as 13, the hidden units in the linear network of the spatial dependence layer and temporal dependence layer are set as 128. The hidden units in the adaptive fusion layer are also set as 128. In the Encoder, the hidden units in the first LSTM layer are set as 256, and 128 in the second layer. Similarly, the hidden units in the LSTM layer for external features are set as 128. For the FNN-Decoder, we set 3 layers of fully connected linear networks. While in the LSTM-Decoder, the first LSTM layer is set as 130, and 256 in the second layer. We use the Adam optimizer to train our model for 100 epochs with an initial learning rate of 0.001.

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TABLE I
EVALUATION METRICS

Forecasting task			Prediction task
MAE	NRMSE	SMAPE	Weighted F1
$\frac{1}{n}\sum_{i=1}^{n}\lvert f_i - y_i\rvert$	$\frac{RMSE}{y_{max}-y_{min}}$	$\frac{100\%}{n}\sum_{i=1}^{n}\frac{\lvert f_i-y_i\rvert}{(\lvert f_i\rvert+\lvert y_i\rvert)^2}$	$\sum_{i=0}^{2} w^i F1$

And for forecasting horizon, we performed ahead predictions for 1-step, 3-step, and 6-step, respectively. For all models, the learning rate is chosen from{0.0001, 0.001, 0.01}, the batch size setting from {64, 128, 256}, and the Dropout from {0, 0.3, 0.5}. The hidden layer size is selected from {64, 128, 256}. The setting of hyperparameters was obtained by several experiments. All comparative experiments are implemented in Python 3.9 on Nvidia GeForce RTX 3090 Ti.

Evaluation Metrics (Table I): for the price forecasting task, we use Mean Absolute Error (MAE), Normalized Root Mean Square Error (NRMSE) and Symmetric Mean Absolute Percentage Error (SMAPE) as the evaluation metrics. For the event prediction task, we use weighted-F1 as the evaluation metrics. Weighted-F1 takes into account the importance of different categories, and the sample size of each category is used as a weight to calculate the weighted-F1, which can be used in an imbalanced dataset. In Table I, f_i and y_i denote the predicted and actual electricity price respectively, n is the test dataset size and i is the index value of test instance, $RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(f_i - y_i)^2}$. w^i is calculated based on the number of samples in each category, $F1 = \frac{2(Precision \times Recall)}{Precision+Recall}$.

A. Baselines

We compare our model with the following 6 baselines for both the price forecasting task and the event prediction task. And 60% of the dataset is used as the training set, 20% as the validation set, and the remaining 20% as the test set. We use the same sliding window size $d = 64$ and use Adam as an optimizer.

For price forecasting task, we use LSTM-CNN, Seq2Seq, Informer [35], ST-Norm [36], MTGNN [37] and ASTGCN [29] as the baselines. Specifically, MTGNN and ASTGCN are GNN-based model.

For spiking event prediction task, we use SVM [38], Randomforest [39], Logistic Regression [40], LSTM-AE [41] and LSTM-MLP-AE as the baselines.

B. Price Forecasting Performance Comparison

The price forecasting comprehensive evaluation results of the MGAAL model and other baselines for different prediction horizons are shown in Table II. The best scores are bolded. It can be seen that the proposed model obtains well prediction performance for most of the evaluation metrics in all prediction horizons with the remaining scores being close to the best.

As seen in Table II, electricity prices in QLD and SA are more difficult to predict. This may be due to the inconsistent distribution of training and test sets in these states, leading to slower improvement in multi-task training. In addition, 65% of SA's electricity generation comes from renewable sources, resulting in more frequent price fluctuations and leading to poor performance by most models. Due to the combination of spatial features, it can be seen that the model based on the GNN framework performs better in the prediction results than Informer and ST-Norm, which indicated the effectiveness of spatial feature derived from GNN. In addition, we compare MGAAL to another GNN-based model, ASTGCN. We considered multi-task on the basis of ASTGCN. It can be seen that our model outperforms ASTGCN in most cases. The visualization results for price forecasting in different states are shown in Fig. 3.

C. Performance Comparison of Event Prediction

The prediction result of price spikes anomaly event with other baselines can be seen in Table III. The best scores are bolded. It can be seen that our model still outperforms the other models. Through this experiment, it can be shown that MGAAL can effectively capture the dynamics of electricity prices behavior by considering the spatiotemporal correlation of time series through graph neural networks, as well as by sharing data and spatiotemporal characteristics of different markets through multi-task learning to predict both prices and spiking events.

D. Ablation Experiment of MGAAL for Price Forecasting Task and Event Predication Task

We analyze the effectiveness of some modules of MGAAL for price forecasting task and event prediction task (Tables IV, and V, the best scores are bolded). The influence of two graph attention layers in our model is examined by disabling the spatial dependence layer and the temporal dependence layer (denoted as w/o spatial dependence, w/o temporal dependence). And we directly concatenate the features together to analyze the effectiveness of the adaptive fusion layer (denoted as w/o adaptive fusion). We also verify the effect of the residual structure by feeding the features directly into the decoders (denoted as w/o residual structure). For studying the role of the intermediate supervisory layer, past value reconstructed loss, and past label reconstructed loss (denoted as w/o supervision, w/o past value reconstructed, w/o past label reconstructed), we compare it with the original model by not taking into account these loss functions in the total loss. And for the effectiveness of multi-task learning (denotes single-task learning), we also compare LSTM-decoders(denoted as LSTM forecasting decoders) with Linear network-decoder. It should be noted that all parameters are set identically. According to the ablation experiments, we found that using the temporal-dependence layer alone leads to an increase in MAE scores. Therefore, the spatial-dependence layer is useful for accurately capturing this correlation because of the power transmission between different power markets. In addition, multi-task learning is also crucial for the final performance.

E. Attention Matrix Interpretability Analysis

To further examine the role of the attention matrix obtained from temporal and spatial attention, we visualized attention by showing the importance of each time and state through heat maps. The temporal-dependence attention can be seen in

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TABLE II
PRICE FORECASTING PERFORMANCE COMPARISON

Step	Model	NSW			QLD			SA			TAS			VIC		
		MAE	NRMSE	SMAPE	MAE	NRMS	SMAPE	MAE	NRMS	SMAPE	MAE	NRMS	SMAPE	MAE	NRMS	SMAPE
1 step	LSTM-CNN	9.0566	0.0284	0.2001	54.5394	0.0364	0.5663	55.8974	0.0311	0.4934	10.6784	0.0544	0.1684	20.593	0.0274	0.3282
	Seq2Seq	9.6567	0.0281	0.122	52.343	0.0363	0.1473	61.1919	0.0315	0.347	10.57	0.0542	0.1775	22.7654	0.0275	0.3114
	Informer	12.8902	0.0296	0.2301	51.0672	0.031	0.47	54.0097	0.0279	0.5858	12.7586	0.0502	0.3595	18.6547	0.0172	0.5751
	ST-Norm	7.8392	0.026	0.119	54.6283	0.0365	0.1373	47.9681	0.0276	0.3275	10.7176	0.0479	0.1879	22.1305	0.0265	0.3116
	MTGNN	8.9058	0.0289	0.1174	40.8353	**0.0243**	**0.1107**	42.6024	0.0246	**0.2858**	10.8577	0.0513	**0.1639**	20.3122	0.0198	0.3216
	ASTGCN	**7.6174**	0.0281	0.1667	52.5057	0.0364	0.2333	47.7067	0.0258	0.5202	11.3052	0.0549	0.3476	22.0721	0.0267	0.4918
	MGAAL	7.705	**0.0248**	**0.1173**	**40.0756**	0.0297	0.2013	**40.0634**	**0.0208**	0.2973	**10.1256**	**0.0453**	0.1814	**17.6166**	**0.0148**	**0.3036**
3 steps	LSTM-CNN	12.0508	0.033	0.191	70.9216	0.0407	0.77	76.7681	0.0347	0.7158	12.9925	0.0572	0.2258	25.8632	0.0275	0.3483
	Seq2Seq	10.1909	0.0303	0.1794	63.547	0.0331	0.284	67.7988	0.032	0.4604	14.8148	0.0575	0.2223	25.3414	0.0276	0.3774
	Informer	14.2502	0.0318	0.2217	54.7359	0.036	0.4662	56.6184	0.0256	0.6156	14.7483	0.0619	0.3812	22.3456	0.0263	0.6149
	ST-Norm	11.5372	0.0305	0.195	50.219	0.0365	0.1685	59.6587	0.0262	0.3952	15.2895	0.0573	0.2534	28.378	0.0252	0.449
	MTGNN	8.5315	0.0297	0.1375	**42.7787**	**0.026**	**0.1427**	50.4545	0.0242	0.3305	**12.0951**	**0.0554**	**0.1971**	25.2234	0.0255	0.3486
	ASTGCN	**8.4936**	0.0301	0.154	55.4566	0.0465	0.2172	49.25	0.036	0.5047	13.12	0.0593	0.305	23.63	0.0268	0.4776
	MGAAL	9.0423	**0.0264**	**0.1361**	46.0867	0.0283	0.2287	**44.8382**	**0.0233**	**0.3288**	12.4856	0.0559	0.2218	**21.7524**	**0.024**	**0.3442**
6 steps	LSTM-CNN	13.6474	0.0387	0.2191	89.597	0.0449	0.6987	80.0155	0.0353	0.769	15.4615	0.0611	0.2442	27.2138	0.0275	0.3979
	Seq2Seq	12.3736	0.0365	0.2687	70.5175	0.04	0.4726	73.9233	0.0338	0.7452	19.5291	0.0654	0.2839	28.8235	0.0276	0.465
	Informer	13.0866	0.0312	0.3067	56.5396	0.0387	0.6223	69.6945	0.0261	0.7325	17.6381	0.0663	0.4789	25.2412	0.027	0.6962
	ST-Norm	10.8681	0.03	0.1476	63.6105	0.0475	0.2041	57.6061	0.0257	0.4585	16.0702	0.0615	0.2593	27.8608	0.0271	0.4262
	MTGNN	9.5234	0.0334	**0.1298**	48.4844	0.0375	**0.1595**	53.7349	0.0341	**0.3684**	**13.9157**	**0.0598**	**0.2254**	26.0532	0.0258	0.3805
	ASTGCN	**9.2805**	0.0329	0.1515	58.0612	0.0466	0.2091	52.02	0.0361	0.4861	14.57	0.0625	0.2965	25.11	0.0269	0.46
	MGAAL	9.7456	**0.0288**	0.1408	**40.1628**	**0.0281**	0.2363	**46.5384**	**0.0246**	0.3711	16.1332	0.0676	0.2506	**22.375**	**0.0251**	**0.3798**

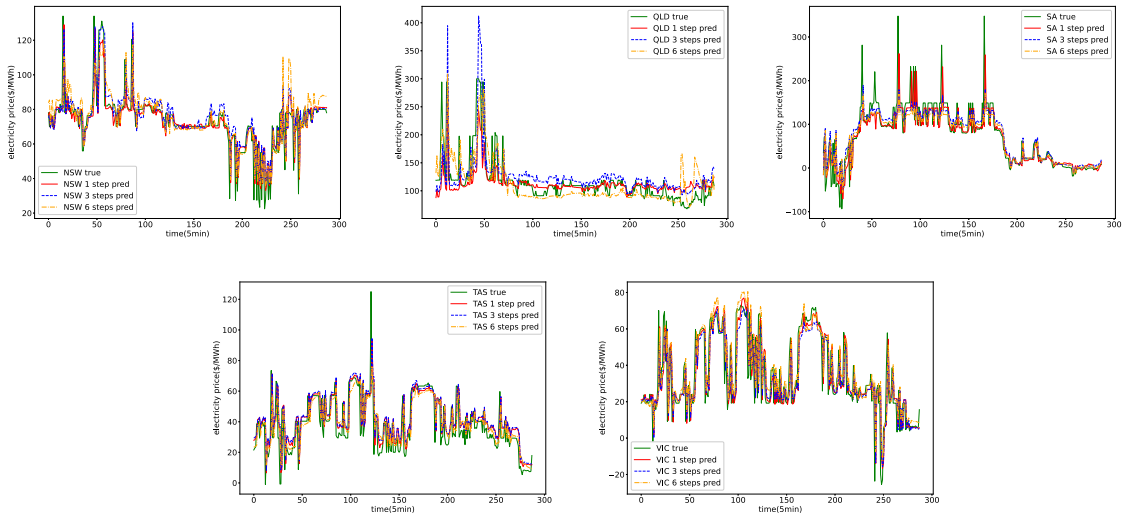


Fig. 3. Visualization results for price prediction. (3a). NSW; (3b). QLD; (3c). SA; (3d). TAS; (3e). VIC;.

TABLE III
EVENT PREDICTION PERFORMANCE COMPARISON

Step	Model	NSW Weighted-F1	QLD Weighted-F1	SA Weighted-F1	TAS Weighted-F1	VIC Weighted-F1
1 step	SVM	0.9	0.81	0.66	0.78	0.8
	Randomforest	0.94	0.88	0.79	0.84	0.82
	LogisticRegression	0.91	0.82	0.72	0.79	0.79
	LSTM-AE	0.94	0.86	0.7	0.82	0.8
	LSTM-MLP-AE	0.95	0.86	0.67	0.82	0.8
	ASTGCN	0.91	0.82	0.6	0.8	0.77
	MGAAL	**0.96**	**0.92**	**0.80**	**0.86**	**0.90**
3 steps	SVM	0.73	0.77	0.5	0.78	0.61
	Randomforest	0.92	0.84	0.73	0.81	0.79
	LogisticRegression	0.91	0.82	0.7	0.78	0.78
	LSTM-AE	0.92	0.84	0.55	0.79	0.78
	LSTM-MLP-AE	0.92	0.82	0.56	0.8	0.76
	ASTGCN	0.91	0.82	0.6	0.79	0.76
	MGAAL	**0.95**	**0.91**	**0.78**	**0.84**	**0.88**
6 steps	SVM	0.88	0.78	0.49	0.74	0.54
	Randomforest	0.9	0.8	0.72	0.79	0.77
	LogisticRegression	0.91	0.81	0.68	0.78	0.78
	LSTM-AE	0.91	0.66	0.5	0.7	0.76
	LSTM-MLP-AE	0.91	0.74	0.47	0.78	0.74
	ASTGCN	0.86	0.71	0.71	0.72	0.65
	MGAAL	**0.94**	**0.91**	**0.77**	**0.83**	**0.86**

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TABLE IV
ABLATION EXPERIMENT OF GAT LAYER, INTERMEDIATE SUPERVISORY LAYER, AND EVENT PREDICTION TASK I

Model	NSW			QLD			SA			TAS			VIC		
	MAE	NRMSE	SMAPE	MAE	NRMS	SMAPE	MAE	NRMS	SMAPE	MAE	NRMS	SMAPE	MAE	NRMS	SMAPE
w/o spatial dependence	7.7841	0.0239	0.1233	41.3379	0.032	0.2387	37.5835	0.0216	0.3733	11.4685	0.0535	0.1876	18.4747	0.0171	0.2968
w/o temporal dependence	8.0983	0.0243	0.12	44.9547	0.0289	0.2236	39.8004	0.0217	0.3592	11.0529	0.0515	0.1838	17.8157	0.0149	0.2865
w/o adptive fussion	7.963	0.025	0.1258	38.6389	0.0282	0.2388	37.2568	0.0214	0.3878	11.2313	0.0536	0.1835	18.893	0.0171	0.2961
w/o residual structure	7.8812	**0.0236**	0.1252	61.1835	0.039	0.1946	46.9605	0.0221	0.4998	11.745	0.0538	0.1815	17.8649	0.0149	0.3254
w/o supervision	7.7596	0.0248	0.1219	39.5006	0.029	0.2461	36.2403	0.0211	0.3799	11.6681	0.0555	0.1793	17.6512	**0.0148**	0.2993
w/o past value reconstructed	7.909	0.025	0.1186	39.7311	0.0292	0.222	41.3466	0.0218	0.4562	11.0327	0.0515	0.1934	17.9447	0.0149	0.3148
w/o past label reconstructed	**7.5539**	0.0248	0.1237	44.2654	0.0303	0.201	36.5624	0.0211	0.3791	11.4982	0.0549	0.1834	18.3503	0.0171	0.2954
LSTM forecast dedcoder	7.6435	0.0231	0.1285	**37.5095**	**0.0288**	**0.1755**	**36.2252**	0.021	0.373	10.1428	0.0506	0.1652	17.7217	**0.0148**	0.2879
single task learning	8.0838	0.0252	0.1309	41.1635	0.0292	0.2581	41.5833	0.0217	0.4023	11.1944	0.053	**0.178**	17.6648	**0.0148**	0.3199
MGAAL	7.705	0.0248	**0.1173**	40.0756	0.0297	0.2013	40.0634	**0.0208**	**0.2973**	**10.1256**	**0.0453**	0.1814	**17.6166**	**0.0148**	**0.2836**

TABLE V
ABLATION EXPERIMENT OF GAT LAYER, INTERMEDIATE SUPERVISORY LAYER, AND EVENT PREDICTION TASK II

Model	NSW Weighted-F1	QLD Weighted-F1	SA Weighted-F1	TAS Weighted-F1	VIC Weighted-F1
w/o spatial dependence	0.95	**0.92**	0.79	0.86	**0.90**
w/o temporal dependence	0.95	**0.92**	0.79	0.87	**0.90**
w/o adptive fussion	0.95	**0.92**	0.79	0.87	**0.90**
w/o residual structure	0.95	0.90	**0.80**	0.80	0.81
w/o supervision	0.95	**0.92**	0.79	0.86	**0.90**
w/o past value reconstructed	**0.96**	**0.92**	**0.80**	**0.88**	**0.90**
w/o past label reconstructed	**0.96**	**0.92**	0.79	0.87	**0.90**
LSTM forecast dedcoder	0.95	**0.92**	**0.80**	**0.88**	0.89
MGAAL	**0.96**	**0.92**	**0.8**	**0.86**	**0.90**

TABLE VI
ABLATION EXPERIMENT OF ADAPTIVE WEIGHTING OPTIMIZATION I

Model	NSW			QLD			SA			TAS			VIC		
	MAE	NRMS	SMAPE	MAE	NRMS	SMAPE	MAE	NRMS	SMAPE	MAE	NRMS	SMAPE	MAE	NRMS	SMAPE
MGAAL Fixed weights	7.9705	**0.0247**	0.1244	45.7926	**0.0287**	0.3568	40.9688	0.0223	0.4551	11.0792	0.0528	0.1885	**16.9896**	**0.0142**	0.3151
MGAAL with awl	**7.6892**	0.0248	0.1186	45.5161	0.0322	0.2631	43.5241	0.021	0.413	13.8912	0.0597	**0.1794**	17.1375	0.0147	0.329
MGAAL	7.705	0.0248	**0.1173**	**40.0756**	0.0297	**0.2013**	**40.0634**	**0.0208**	**0.2973**	**10.1256**	**0.0453**	0.1814	17.6166	0.0148	**0.3036**

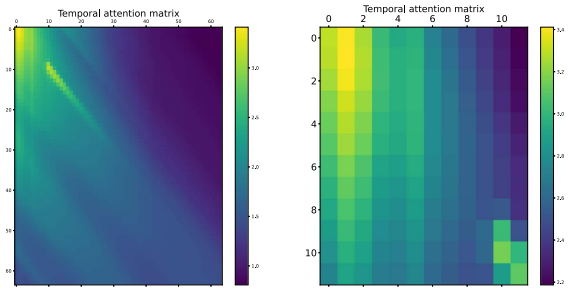


Fig. 4. Visualization results for Temporal dependence layer where 4b is an enlarged version in 4a.

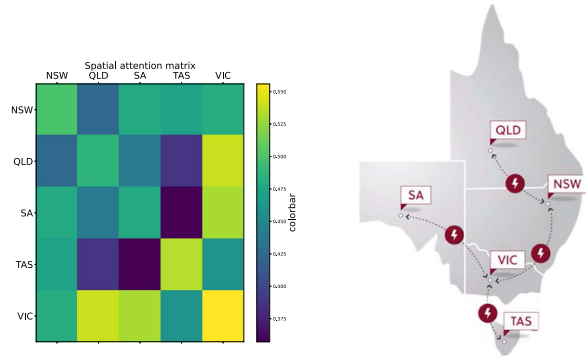


Fig. 5. Visualization results for Spatial dependence layer, Where 5b denotes the topology of National Electricity Market of Australia [2].

Fig. 4, where the X-axis and Y-axis represent the time series in the sliding window. As time progresses, the attention to later moments tends to decrease, but later moments still have some attention weight towards earlier moments. The attention weight on the lower left side is larger than that on the upper right side, which aligns with reality because historical moments have some influence on the present, but the future is not known to past moments, leading to a small attention weight between them. The spatial-dependence attention can be seen in Fig. 5(a). In the spatial attention matrix, VIC has a larger attention weight than other states, which can be explained by the fact that VIC has more neighboring nodes in Fig. 5(b). Therefore, the spatial-dependence layer and temporal-dependence layer in MGAAL can effectively capture the spatiotemporal characteristics of different power markets.

TABLE VII
ABLATION EXPERIMENT OF ADAPTIVE WEIGHTING OPTIMIZATION II

Model	NSW Weighted-F1	QLD Weighted-F1	SA Weighted-F1	TAS Weighted-F1	VIC Weighted-F1
Fixed weights	0.95	**0.92**	0.79	0.84	0.89
MGAAL with awl	0.95	**0.92**	**0.81**	0.79	0.89
MGAAL	**0.96**	**0.92**	0.80	**0.86**	**0.90**

F. Weight Assignment Methods Performance Comparison

We test the effectiveness of different weight assignment methods on multi-task learning. The result can be seen in Tables VI and VII (the best scores are bolded). Among that, the

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TABLE VIII
PRICE FORECASTING PERFORMANCE COMPARISON IN DIFFERENT TIME PERIOD (6 STEPS)

Step	Model	NSW			QLD			SA			TAS			VIC		
		MAE	NRMSE	SMAPE	MAE	NRMS	SMAPE	MAE	NRMS	SMAPE	MAE	NRMS	SMAPE	MAE	NRMS	SMAPE
Middle	LSTM-CNN	26.1212	0.0339	0.331	42.8626	0.0684	0.7234	41.897	0.1142	0.7593	16.0782	0.0691	0.7541	29.1028	0.1362	0.5627
	Seq2Seq	29.2207	0.0341	0.4037	47.6977	0.0687	0.7779	38.2052	0.1121	0.7007	16.3979	0.07	0.7558	27.9935	0.1312	0.6025
	Informer	29.5527	0.0363	0.4326	49.1532	0.0718	0.8655	41.1532	0.0918	0.8655	15.9238	0.0698	0.776	27.6573	0.1049	0.5821
	ST-Norm	30.711	0.0351	0.4223	39.4973	0.0672	0.5589	38.8027	0.1166	0.7318	16.9877	0.0726	0.7712	35.6485	0.1498	0.7005
	MTGNN	**19.5047**	**0.0325**	**0.3126**	59.1028	**0.0665**	**0.5509**	52.4026	0.0355	0.4323	**12.2532**	**0.0526**	0.699	28.9595	0.1365	0.5929
	ASTGCN	28.8861	0.0351	0.3913	39.9548	0.0674	0.5641	29.5355	0.0353	0.405	16.8491	0.0687	0.7805	29.0737	0.1351	0.5951
	MGAAL	25.7904	0.035	0.3296	**39.4862**	0.0676	0.5964	**29.5309**	**0.0325**	**0.4026**	15.8929	0.0684	**0.6776**	**26.6798**	**0.1311**	**0.5621**
Winter	LSTM-CNN	40.5423	0.0963	1.0621	38.4658	0.173	0.5387	51.4236	0.1456	1.0156	18.5844	0.117	0.7785	69.5735	0.2621	1.0302
	Seq2Seq	33.8169	0.1175	0.5378	42.4049	0.1485	0.6967	48.071	0.1054	1.001	18.5215	0.1169	0.7576	35.9815	0.1192	1.0111
	Informer	43.7372	0.1432	0.6348	52.3067	0.211	0.9231	46.2985	0.0998	1.0465	30.8613	0.1768	00.9226	41.4721	0.1156	0.9842
	ST-Norm	44.9095	0.1064	0.9426	42.3119	0.1464	0.966	52.7247	0.1076	1.2382	18.7101	0.1166	0.8852	46.2787	0.1253	1.1413
	MTGNN	**30.0512**	**0.0927**	**0.414**	44.1338	0.1443	**0.4487**	45.177	0.0849	0.8351	**11.5463**	0.052	0.875	**30.0403**	**0.0861**	0.7984
	ASTGCN	30.1639	0.1189	0.5111	38.4467	0.1426	0.6306	38.3027	0.1183	0.8209	30.3023	0.1603	0.8109	30.3139	0.096	0.7931
	MGAAL	31.6884	0.1236	0.1236	**38.2107**	**0.1424**	0.8994	**35.3823**	**0.0822**	0.818	18.4534	**0.1159**	**0.7548**	32.4662	0.0947	**0.7911**

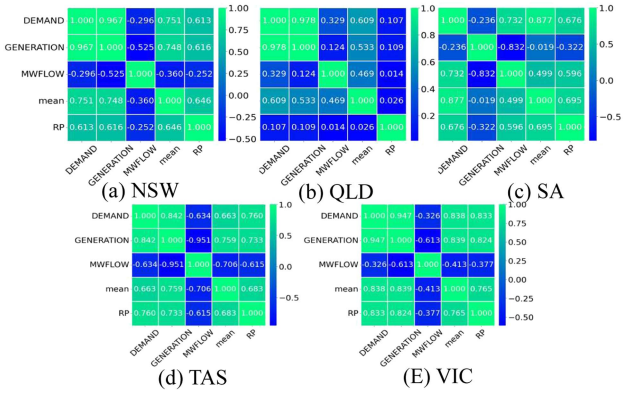


Fig. 6. Correlation coefficients between external features and electricity price.

'Fixed weights' represents the weights selected by hand through multiple experiments, 'awl' denotes the method proposed in paper [34]. It can be seen that the optimized adaptive weight assignment method in our model works better than the fixed weight and adaptive weight assignment in this scenario.

G. External Feature Analysis

In MGAAL, temporal features, electricity demand, generation, average price, and power flow serve as external features, exerting a substantial impact on our model's predictive capabilities. Based on an examination of the data, electricity prices are lower on weekends and higher at night, making price spikes more likely on weekdays or evenings. Additionally, when demand exceeds generation capacity, it can lead to scarcity of electricity, increasing competition and potentially causing price spikes. Sudden shortages of external power supply at times of high demand can also trigger price spikes. We visualize the correlation between external features and electricity prices through heatmaps (Fig. 6), thus showing the importance of external features on electricity prices. As shown in the heatmaps, the electricity demand (DEMAND), generation (GENERATION), external inputs (MWFLOW) and the 28-day average of the electricity prices (mean) correlate with the price of electricity (RP). Therefore, inputting these features simultaneously will help improve the accuracy of abnormal price spikes and price forecasting.

H. Robustness Experiment With Different Time Periods

In the above section, we have tested the algorithm using data collected from AEMO from October 2021 to January 2022. Since electricity prices fluctuate over time, it is important to test the robustness of the algorithm. We incorporated data from different times for experimentation. According to the paper [17], the analysis of electricity price spikes in Australia is conducted considering three types of seasons: Winter (May–Aug.), Middle (Mar.–Apr., Sep.–Oct.), and Summer (Nov.–Feb.). In our study, we construct two datasets for the Middle and Winter seasons from the NEM website [2]. Due to differences in dataset resolution, with a half-hour resolution before October and a 5-minute resolution after, we excluded October (already been used in previous experiments) and selected March and April for the Middle season dataset. The dataset of the Middle season comprises a total of 14,640 data points and spike occurrences in electricity prices for various states are as follows: NSW(11.65%), QLD(10.79%), SA(5.81%), TAS(7.04%), VIC(10.08%). While in the Winter season dataset, there are 29,520 data points and spike occurrences in electricity prices for various states are as follows: NSW(10.13%), QLD(9.11%), SA(7.5%), TAS(6.32%), VIC(8.93%). Dataset segmentation and model parameters remained consistent with previous settings. We employ the same baseline models as the reference for the 6-step electricity price forecasting task and abnormal price spiking event forecasting task in different time period datasets. The results are presented in Tables VIII and IX, the best scores are bolded. For forecasting error, our model outperforms better than other baselines in most cases. Specifically, in the Middle dataset results, when compared to MTGNN, the average MAE and SMAPE have decreased by 22.6% and 2.7% respectively. Similarly, in the Winter dataset tests, our model has achieved a 3.0% decrease in average MAE compared to MTGNN. As seen in the results, TAS exhibits lower performance compared to MTGNN, possibly attributed to its lower rate of spike occurrences, making it more challenging to predict the label accurately. Therefore, in multitask optimization, the loss focuses on optimizing the prediction of anomalous price events, resulting in a slower decrease in loss for the price prediction task. For label forecasting, since there are fewer anomalous electricity price spike rates in the Middle and Winter seasons datasets for QLD and SA compared to October-January (peak season), the prediction errors are also relatively smaller.

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TABLE IX
EVENT PREDICTION PERFORMANCE COMPARISON IN DIFFERENT TIME PERIODS (6 STEPS)

Step	Model	NSW Weighted-F1	QLD Weighted-F1	SA Weighted-F1	TAS Weighted-F1	VIC Weighted-F1
Middle	SVM	**0.86**	**0.86**	0.7	0.64	0.64
	Randomforest	0.83	0.85	**0.83**	0.72	0.77
	LogisticRegression	0.78	0.85	0.79	0.74	**0.78**
	LSTM-AE	0.75	0.76	0.68	0.74	0.64
	LSTM-MLP-AE	0.74	0.71	0.78	0.73	0.66
	ASTGCN	0.75	0.64	0.73	0.73	0.74
	MGAAL	0.8	**0.86**	0.77	**0.75**	**0.75**
Winter	SVM	**0.93**	**0.93**	**0.93**	0.92	0.93
	Randomforest	**0.93**	**0.93**	**0.93**	0.92	0.95
	LogisticRegression	0.91	0.92	0.92	0.92	0.94
	LSTM-AE	0.92	0.91	0.92	0.94	0.92
	LSTM-MLP-AE	0.89	0.9	0.9	0.91	0.93
	ASTGCN	0.91	0.92	0.91	0.91	0.92
	MGAAL	0.92	0.91	**0.93**	**0.93**	**0.96**

Additionally, the Middle season dataset, being smaller, is less accurate in predicting anomalous price spike events compared to the other two datasets.

V. CONCLUSION

Electricity price forecasting task is complex due to its nonlinear and volatile nature. Our research introduces MGAAL, a novel deep learning framework for multivariate multi-step electricity price forecasting. It incorporates an auxiliary task to predict abnormal price spikes, leveraging spatiotemporal correlations on a graph. MGAAL uses multiple decoders and adaptive task weights for better forecasting, validated on Australian market data. However, the paper lacks an in-depth interpretability study for electricity price anomalies and relies on supervised learning, potentially introducing labeling errors. Therefore, in the future, it can be worthwhile to consider self-supervised learning and increase the interpretability study or further exploration in methodology for spike price modeling.

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